

### 3.2.3 Chemical equilibrium

| Learning outcomes                                                                                                                                                                                                      | Additional guidance                                                                                                                                                       |
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| <i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>                                                                                                                          |                                                                                                                                                                           |
| <b>Dynamic equilibrium and le Chatelier's principle</b>                                                                                                                                                                |                                                                                                                                                                           |
| (a) explanation that a dynamic equilibrium exists in a closed system when the rate of the forward reaction is equal to the rate of the reverse reaction and the concentrations of reactants and products do not change |                                                                                                                                                                           |
| (b) le Chatelier's principle and its application for homogeneous equilibria to deduce qualitatively the effect of a change in temperature, pressure or concentration on the position of equilibrium                    | Definition for le Chatelier's principle <b>not</b> required.<br><br>HSW1,2,5 Use of le Chatelier's principle to explain effect of factors on the position of equilibrium. |
| (c) explanation that a catalyst increases the rate of both forward and reverse reactions in an equilibrium by the same amount resulting in an unchanged position of equilibrium                                        |                                                                                                                                                                           |
| (d) the techniques and procedures used to investigate changes to the position of equilibrium for changes in concentration and temperature.                                                                             | Qualitative effects only.<br><br>HSW4 Opportunities to carry out experimental and investigative work.                                                                     |
| (e) explanation of the importance to the chemical industry of a compromise between chemical equilibrium and reaction rate in deciding the operational conditions                                                       | HSW6 Balancing the effects of equilibrium, rate, safety and economics to determine the conditions used in industrial reactions e.g. Haber process.                        |
| <b>The equilibrium constant, <math>K_c</math></b>                                                                                                                                                                      |                                                                                                                                                                           |
| (f) expressions for the equilibrium constant, $K_c$ , for homogeneous reactions and calculations of the equilibrium constant, $K_c$ , from provided equilibrium concentrations                                         | $M0.2$ , $M1.1$ , $M2.3$ , $M2.4$<br><br>Learners will <b>not</b> need to determine the units for $K_c$ .                                                                 |
| (g) estimation of the position of equilibrium from the magnitude of $K_c$ .                                                                                                                                            | $M0.3$<br><br>A qualitative estimation only is required.                                                                                                                  |